

CSE461

Introduction to Robotics Lab

Real-World Implication

Group : 01

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Group members :

Iftequr Ahmed	22299172
MD. Nafizul Sadik	21201313

Fatin Rahman Taseen	22299002
Sadat Hossain	22201314
Shafi Ahmed Adib	23101307

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Submitted to - Utsha Kumar Roy and Shreya Adrita Banik

1. **Introduction**: Dengue is one of the biggest public health threats in Bangladesh. Every monsoon season, thousands of people get infected with dengue fever. The hospitals get overwhelmed with patients and the families lose money and lives. The main reason that this outbreak keeps happening is simple and everyone knows about it. Almost all of the buildings in Dhaka city have stagnant water in rooftops, flower pots, and discarded containers. These situations go unnoticed until it is too late. In 2023, Bangladesh recorded 321,179 dengue cases and 1,705 deaths, the highest ever in the country's history (Directorate General of Drug Administration DGDA, 2023). Dhaka alone accounted for over 40% of those cases (Institute of Epidemiology, Disease Control and Research, 2023). Our robot addresses this by autonomously patrolling outdoor spaces and detecting stagnant water using moisture, humidity, and IR sensors and also alerts residents before mosquitoes even get a chance to breed.

2. **Purpose**: This robot exists to fill the gap between government inspections. Health workers cannot visit every household regularly, and most residents do not check their rooftops or corner containers on their own. This robot does it automatically. Its core goals are to detect stagnant water through multi sensor confirmation, patrol areas without human supervision, alert residents immediately upon detection, and provide a solution affordable enough for typical urban households in Bangladesh.

3. **Societal implications**: Firstly, dengue has impacted almost every family whether they have high or low income.However, the most unhygienic places in Dhaka city are the places where most of the low income families reside. Dengue impacts these families who cannot afford pest control services and rarely receive government inspections. Studies show that only about 20% of urban households in Bangladesh receive any form of regular dengue prevention visits (DGDA, 2022). Our robot solves this problem with a little workshop on its usage. There is also a behavioral side to this. When a robot detects a spot a resident never noticed, it changes how that person thinks about water management. That shift in awareness is a long term societal benefit.

4. **Health implications**: According to WHO (2021), Eliminating breeding spots before larvae develop is the most effective dengue control strategy available. Each breeding spot removed can prevent an estimated 100–500 mosquito larvae from reaching adulthood. Dengue not only just causes fever. Dengue hemorrhagic fever (DHF) and dengue shock syndrome (DSS) can be fatal if medical steps are not taken timely. There is also an indirect health benefit worth mentioning. During peak dengue season, hospitals in Dhaka get severely overcrowded. It spreads dengue even more and makes the job more difficult for doctors. So, preventing

dengue is a must in order to solve this problem and our robot is equipped with the necessary tools to tackle this.

5. **Safety implications**: The robot is slow moving and low to the ground. The physical risk to people is minimal. Practical risk is more of our concern as the body of the robot is not that strong. Rolling off a ledge is also a concern but it will be sorted with future upgrades.

6. **Legal implications**: Bangladesh does not yet have dedicated robotics legislation, but existing laws apply. The Digital Security Act (2018) governs any stored data, so if the robot logs location data on an SD card, access to that data needs to be restricted and managed responsibly. On the property side, deploying the robot in shared apartment rooftops or courtyards will require consent from all relevant residents or building management.

7. **Cultural implications:**

   In our country, technologies are not widely used, as many of our people, especially older and rural citizens, rely heavily on age old traditional methods across many sectors. So user friendly demonstrations and what the robots actually do will go a long way. In more conservative households like ours, any device moving through a private compound needs to be fully controlled by the household owner. The robot's current design, which has no camera and requires manual is well suited for this context. Easy Bengali UI can make it easy for most of our people.

8. **Conclusion:**

Dengue is a disaster in Bangladesh. Our cities are not clean, making them a perfect breeding spot for mosquitoes. This robot focuses on an approach to find stagnant water, raise an alert, and let people act for the prevention. It does not try to solve everything, and it does not need to. With proper weatherproofing, edge safety, and community level awareness, it might move from a university prototype to something that makes a difference in urban neighborhoods across Bangladesh.

References

Directorate General of Drug Administration (DGDA). (2023). *Dengue surveillance data 2023*. Ministry of Health and Family Welfare, Bangladesh.

Directorate General of Drug Administration (DGDA). (2022). *Annual dengue situation report 2022*. Ministry of Health and Family Welfare, Bangladesh.

Institute of Epidemiology, Disease Control and Research (IEDCR). (2023). *Dengue situation report: Bangladesh 2023*. Ministry of Health and Family Welfare, Bangladesh

World Health Organization (WHO). (2021). *Global vector control response 2017–2030*. WHO Press. https://www.who.int/publications/i/item/9789241512978